Assignment No.3 2413FEB2T1 : Engineering Mathematics-2
Topic: Numerical Solution to Ordinary Differential Equations

1 Problems on Taylor Series Method

1. Solve $\frac{dy}{dx} = x^2 - y$ with $y(0) = 1$ at $x = 0.1$ using Taylor Series Method.
2. Solve $\frac{dy}{dx} = 2y + 3e^x$ with $y_0 = 0$ when $x_0 = 0$ at $x = 0.1$ and $x = 0.2$ using Taylor Series Method.
3. Solve $\frac{dy}{dx} = 1 + y^2$ and $y_0 = 0$ when $x_0 = 0$ at $x = 0.2$ and $x = 0.4$ using Taylor Series Method.
4. Find the value of $y(0.2)$ and $y(0.4)$ where $\frac{dy}{dx} = 1 - 2xy$ with $y(0) = 0$ using Taylor Series Method.
5. Solve $\frac{dy}{dx} = x + y^2$ with $y(0) = 1$ at $x = 0.2$ using Taylor Series Method.

2 Problems on Euler's Method

1. Find the approximate value of y at $x = 1$ in five steps if $\frac{dy}{dx} = x + y$ with $y(0) = 1$, taking $h = 0.2$.
2. Find the approximate value of y at $x = 0.3$ if $\frac{dy}{dx} = xy$ with $y(0) = 1$ and $h = 0.1$.
3. Find the approximate value of y at $x = 1.5$ in five steps if $\frac{dy}{dx} = \frac{y - x}{\sqrt{xy}}$ with $y(1) = 2$, taking $h = 0.1$.
4. Find the approximate value of y at $x = 0.4$ with $h = 0.1$ if $\frac{dy}{dx} = 1 + xy$ with $y(0) = 0.2$.
5. Using Euler's Method find the approximate value of y at $x = 0.1$ in 5 steps where $\frac{dy}{dx} = \frac{y - x}{y + x}$ and $y(0) = 1$.

3 Problems on Euler's Modified Method

1. Solve $\frac{dy}{dx} = x^2 + y$ with $y(0) = 1$ and $h = 0.05$ at $x = 0.1$.
2. Solve $\frac{dy}{dx} = x + 3y$ with $x_0 = 0$ and $y_0 = 1$ at $x = 0.1$ in one step.
3. Solve $\frac{dy}{dx} = log(x + y)$ with $y(1) = 2$ at $x = 1.4$ in two steps.

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4. Solve $\frac{dy}{dx} = x - y^2$ with $y(0) = 1$ at $x = 0.4$ in two steps.

5. Solve $\frac{dy}{dx} = 2 + \sqrt{xy}$ with $y(1.2) = 1.6403$ at $x = 1.6$ in two steps correct up to 4 decimal places.

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